Normal Distribution

ANSWER KEY

Section 1: The Standard Normal Distribution $Z \sim N(0,1)$

1 Find $P(Z < 1.645)$ for the standard normal distribution $Z \sim N(0,1)$.

a. 0.9500 ✓

b. 0.9750

c. 0.9000

d. 0.9900

From the standard normal cumulative distribution tables, $\Phi(1.645) = 0.9500$.

2 Find $P(Z > -1.282)$ for $Z \sim N(0,1)$.

a. 0.9000 ✓

b. 0.1000

c. 0.8000

d. 0.8500

By symmetry, $P(Z > -1.282) = P(Z < 1.282) = \Phi(1.282) = 0.9000$.

3 Find $P(-1.96 < Z < 1.96)$ for $Z \sim N(0,1)$.

a. 0.9500 ✓

b. 0.9750

c. 0.9000

d. 0.9900

$P(-1.96 < Z < 1.96) = 2\Phi(1.96) - 1 = 2(0.9750) - 1 = 1.95 - 1 = 0.9500$.

Section 2: Standardization & General Normal Distribution

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4 The random variable $X \sim N(50, 16)$. Find $P(X < 56)$.

a. 0.9332 ✓

b. 0.8413

c. 0.9772

d. 0.8849

Standard deviation $\sigma = \sqrt{16} = 4$. $z = \frac{56 - 50}{4} = 1.5$. $P(X < 56) = \Phi(1.5) = 0.9332$.

5 Heights of adult females are normally distributed with mean 165 cm and standard deviation 6 cm. What percentage of females are taller than 177 cm?

a. 2.28% ✓

b. 5.00%

c. 4.56%

d. 1.14%

$z = \frac{177 - 165}{6} = 2.0$. $P(X > 177) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228 = 2.28\%$.

6 If $Y \sim N(100, 225)$, find $P(85 < Y < 115)$.

a. 0.6826 ✓

b. 0.9544

c. 0.5000

d. 0.7500

$\sigma = \sqrt{225} = 15$. The interval $85 < Y < 115$ represents $\mu \pm 1\sigma$.
$P(-1 < Z < 1) = 2\Phi(1) - 1 = 2(0.8413) - 1 = 0.6826$.

Section 3: Inverse Normal & Finding μ and σ

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7 The random variable $X \sim N(\mu, 25)$ has $P(X < 62) = 0.8413$. Find the mean μ.

a. **57** ✓

b. 55

c. 52

d. 60

$\Phi(z) = 0.8413 \Rightarrow z = 1.0$. Since $\sigma = 5$: $1.0 = \frac{62 - \mu}{5} \Rightarrow 62 - \mu = 5 \Rightarrow \mu = 57$.

8 For a normal variable X, $P(X < 40) = 0.05$ and $P(X < 70) = 0.95$. Find μ and σ.

a. $\mu = 55, \sigma = 9.12$ ✓

b. $\mu = 55, \sigma = 15.0$

c. $\mu = 50, \sigma = 10.0$

d. $\mu = 55, \sigma = 5.50$

By symmetry, $\mu = (40 + 70)/2 = 55$. For 95th percentile, $z = 1.645$: $1.645 = \frac{70 - 55}{\sigma} \Rightarrow \sigma = \frac{15}{1.645} \approx 9.12$.

9 Packets of biscuits have nominal weight 200 g. Actual weights are normally distributed with $\mu = 204$ g and $\sigma = 2.5$ g. What proportion of packets are underweight (< 200 g)?

a. 0.0548 ✓

b. 0.0228

c. 0.0100

d. 0.0808

$z = \frac{200 - 204}{2.5} = -1.6$. $P(X < 200) = \Phi(-1.6) = 1 - \Phi(1.6) = 1 - 0.9452 = 0.0548$.

Section 4: Normal Approximation to the Binomial Distribution

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10 State the two conditions required to approximate a binomial distribution $X \sim B(n, p)$ by a normal distribution.

a. $np > 5$ **and** $n(1-p) > 5$ ✓

b. $n > 30$ and $p = 0.5$

c. $np > 10$ only

d. $n(1-p) > 100$

The standard Cambridge AS Level guideline requires both $np > 5$ and $n(1-p) > 5$ for an adequate normal approximation.

11 A fair coin is tossed 100 times. Using the normal approximation with continuity correction, what is the expression for $P(X \geq 55)$?

a. $P(Y > 54.5)$ **where** $Y \sim N(50, 25)$ ✓

b. $P(Y > 55.5)$ where $Y \sim N(50, 25)$

c. $P(Y > 54.5)$ where $Y \sim N(50, 5)$

d. $P(Y > 55.0)$ where $Y \sim N(50, 25)$

$\mu = 100(0.5) = 50$, $\sigma^2 = 100(0.5)(0.5) = 25$. For $X \geq 55$, the continuity correction shifts to the left boundary: $P(Y > 54.5)$.

12 Compute the approximate probability of getting at least 55 heads in 100 tosses using $Y \sim N(50, 25)$.

a. 0.1841 ✓

b. 0.1587

c. 0.2119

d. 0.1250

$z = \frac{54.5 - 50}{5} = 0.90$. $P(Y > 54.5) = 1 - \Phi(0.90) = 1 - 0.8159 = 0.1841$.

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